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SML ASSIGNMENT - 4
REPORT

Question 1: Implementation of AdaBoost.M1 using Decision Trees

Objective:
The aim is to implement the AdaBoost.M1 algorithm with decision stumps as weak learners, using the MNIST dataset. We are focusing only on the digits '0' and '1', labeled as -1 and 1 respectively.

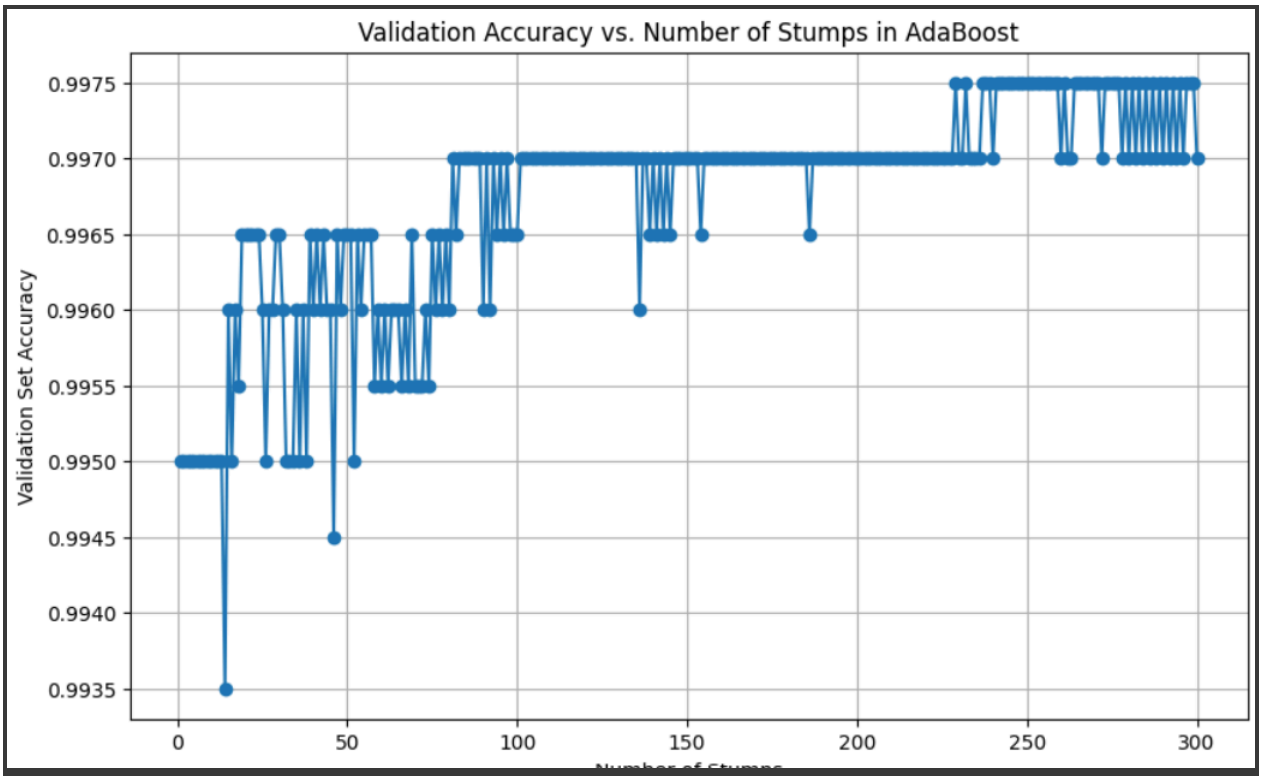
Data Preparation:
Dataset: We use the MNIST dataset, which is split initially into a training set and a test set.
Validation Set: From the training dataset, 1000 samples from each class (0 and 1) are separated to form the validation set. The remaining samples form the new training set.
PCA Application: Principal Component Analysis (PCA) is used to reduce the dimensionality of the training data to 5 dimensions. This reduced dimensionality is supposed to help in simplifying the model without losing significant information.

AdaBoost.M1 Algorithm:
Initialization: Weights are initially set equally for all samples in the training set.
For each of 300 iterations:
- Decision Stump Training: A decision stump is trained as follows:
  - For each dimension, sort the unique values.
  - Evaluate possible splits, which are the midpoints between consecutive unique values.
  - The best split is chosen based on the one that minimizes the weighted misclassification error.
- Weight Update:
  - Compute the error rate of the stump.

- Compute alpha, the weight of the stump, using the formula

$$\alpha_t = \frac{1}{2} \ln \left(\frac{1-\text{error}}{\text{error}} \right)$$

- Update the weights of the samples for the next iteration, emphasizing more on incorrectly classified samples.
- Evaluate on Validation Set: After adding each tree, the combined classifier's performance is evaluated on the validation set.



- Accuracy Tracking: Track and plot accuracy as a function of the number of trees.

Final Model Selection and Testing:
- The decision stump that gives the highest accuracy on the validation set is selected.
- This model is then tested on the test set to report the final test accuracy.

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Test Accuracy: 0.9887
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Question 2: Gradient Boosting for Regression

Objective:
Adapt the problem of classifying digits '0' and '1' into a regression problem using gradient boosting with decision stumps as the weak learners.

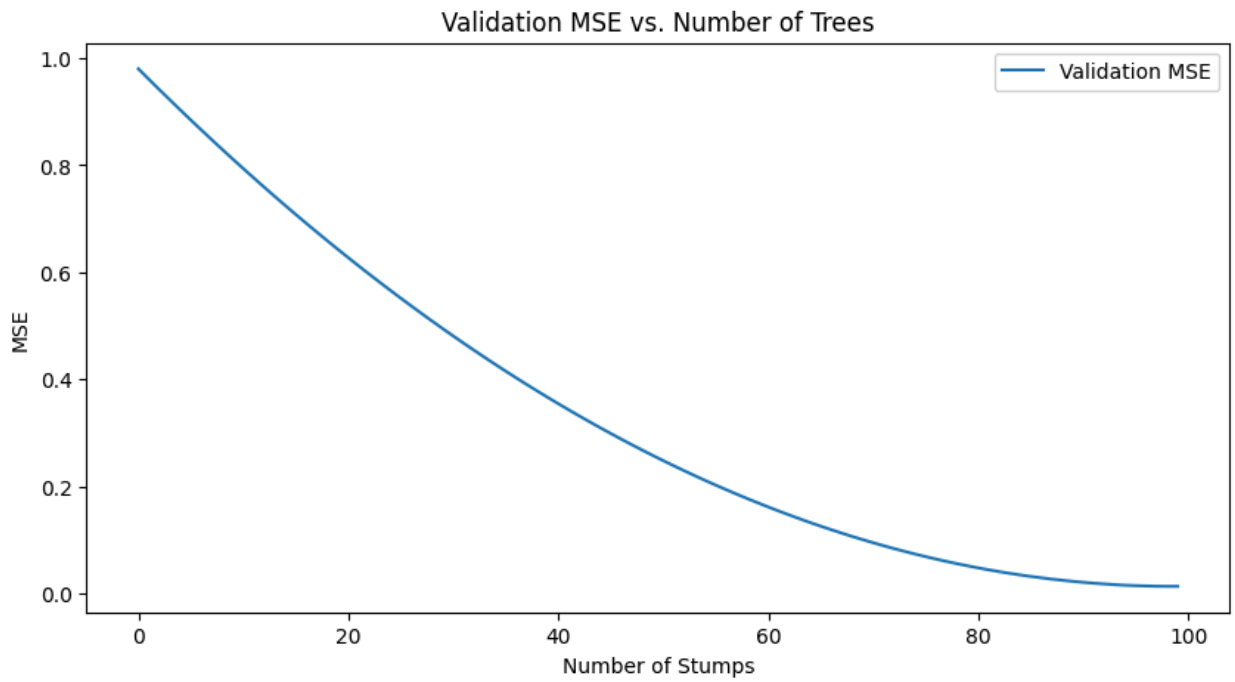
Data Preparation:
- The same data splitting and dimensionality reduction steps as in Question 1 are followed.

Gradient Boosting Algorithm:
-Initialization: Start with an initial model (often zero), and initial residuals are set as the labels.
- For each of 300 iterations:
  -Decision Stump Training:
    - Similar to AdaBoost, but the splits are chosen to minimize the Sum of Squared Residuals (SSR).
    - Find the best split and train the stump to approximate the negative gradient of the loss function (absolute loss).
  - Residue Update:
    - After training each stump, compute the new residuals by subtracting the predictions scaled by a learning rate (e.g., 0.01) from the previous residuals.
  - Evaluate on Validation Set: The Mean Squared Error (MSE) between the predicted values and the actual labels is calculated for the validation set.
  - MSE Tracking: Track and plot MSE as a function of the number of trees.

Final Model Selection and Testing:
- The decision stump that gives the lowest MSE on the validation set is selected.
- This model is then tested on the test set to report the final test MSE.

Conclusion:
Both algorithms, AdaBoost.M1 for classification and gradient boosting for regression, demonstrate how ensemble methods can enhance the performance of simple models like decision stumps by focusing iteratively on difficult to predict

samples or residuals. The use of PCA helps in reducing the complexity and improving the speed of training without compromising on the model's ability to make accurate predictions. The continuous evaluation on a validation set provides a guide to select the most effective model before final testing.



Plot for 100 Stumps

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Final MSE after 300 stumps: 0.012881799999999992
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